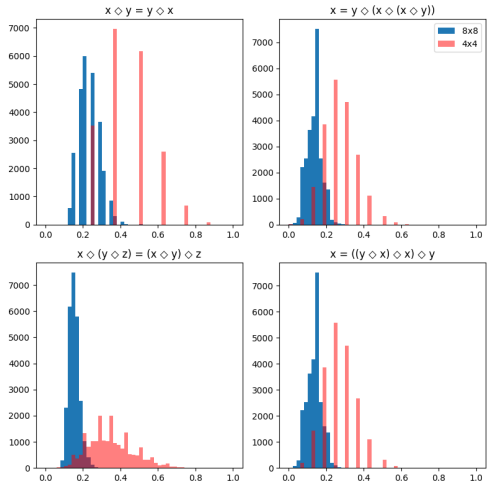

Figure 2: Stone spectra of the commutative law (top-left) and of the associative law (bottom-left) for a sampling of $n = 20\,000$ magmas of size 4 (red curve) and of $n = 24\,000$ magmas of size 8 (blue curve). The Stone spectra of the conjugate theories Eqn[63] and Eqn[271] are identical (top and bottom, right).

$$\circledast \quad : \quad S_n \times \mathbb{H}_n \longrightarrow \mathbb{H}_n$$

of the symmetry group S_n on the n dimensions (or directions) of the space $\mathbb{H}_n$, defined as

$$\sigma \circledast (p_1, \ldots, p_n) := (p_{\sigma(1)}, \ldots, p_{\sigma(n)})$$

for $\sigma \in S_n$ and $\mathbf{p} = (p_1, \ldots, p_n) \in \mathbb{H}_n$. Since the sampling order of the magmas $A_1, \ldots, A_n$ does not matter, it makes sense to consider the position of each equational theory

$$\mathbf{p} := (p_1, \ldots, p_n) \in \mathbb{H}_n$$

modulo the action of the symmetry group S_n. One obtains in this way a position in the quotient space $\mathbb{H}_n/S_n$, defined as the multiset

$$\mathbf{p}\,[S_n] \;:=\; \{\!|\, p_1 \,,\, \ldots \,,\, p_n \,|\!\}$$

We call this multiset the *Stone spectrum* of the equational theory. The Stone spectrum can be nicely visualized as a finite positive measure of weight n on the interval $[0, 1]$, defined as the sum

$$\textstyle\sum_{i=1}^{n} \delta_{p_i}$$

of the Dirac distributions δ_{p_i}, see Fig. 2 for an illustration.

5.4 Stone interference spectrum

The cartesian product of the hypercube $\mathbb{H}_n = [0, 1]^n$ with itself comes equipped with an action

$$\circledast \quad : \quad S_n \times \mathbb{H}_n \times \mathbb{H}_n \longrightarrow \mathbb{H}_{\times}\mathbb{H}_n$$

defined componentwise:

$$\sigma \circledast (\mathbf{p}, \mathbf{q}) \;:=\; (\sigma \circledast \mathbf{p}, \sigma \circledast \mathbf{q})$$

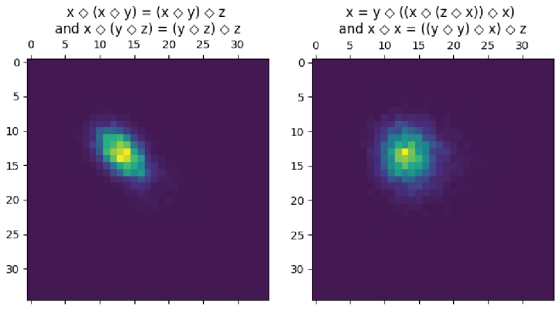

Figure 3: The Stone interference spectrum of Eqn[4400] and Eqn[4533] detects a degree of statistical dependence between the two conjugate equational theories (left) while the Stone interference spectrum of Eqn[1092] and Eqn[4092] shows full statistical independence between the two equational theories (right).

Instead of looking at the relative positions of a pair of equational theories Eqn[j] and Eqn[k] in the feature space $\mathbb{H}_n \times \mathbb{H}_n$, we may look at their relative positions in the quotient space $\mathbb{H}_n \times \mathbb{H}_n / S_n$. This is provided by the finite multiset of pairs:

$$(\mathbf{p}, \mathbf{q})\,[S_n] \;:=\; \{\!|\, (p_1, q_1) \,,\, \ldots \,,\, (p_n, q_n) \,|\!\}$$

which we call *Stone interference spectrum*. In the same way as the Stone spectrum of an equational theory, this interference spectrum of two equational theories can be described as the finite measure on the square $[0, 1]^2$ defined as the sum

$$\textstyle\sum_{i=1}^{n} \sum_{j=1}^{n} \delta_{(p_i, q_j)}$$

where $\delta_{(p,q)}$ denotes the Dirac distribution of $(p, q) \in [0, 1]^2$. This representation enables one to visualize the Stone interference spectrum in an instructive way, see Fig. 2 for two illustrations.

6 The latent space of equational theories

6.1 Building the latent space using a PCA dimensionality reduction

We apply a principal component analysis (PCA) method in order to derive a 3-dimensional representation of the 4694 equational theories from thr positions in the n-dimensional feature space $\mathbb{H}_n$. We thus start by computing the center of gravity $\mu \in \mathbb{H}_n$ of the 4694 vertices in $\mathbb{H}_n$. We subtract μ from each column of the matrix R defined in (1), in order to obtain a matrix $X = R - \mu$ with zero-mean rows. The PCA method then diagonalizes the covariance matrix $C_X = \frac{1}{n} X X^T$, see (Shlens, 2014; Hanchi et al., 2025) for details.

We keep the first three principal components or dimensions (noted X, Y, Z) of the row-space which we call the *latent space of equational theories*. This

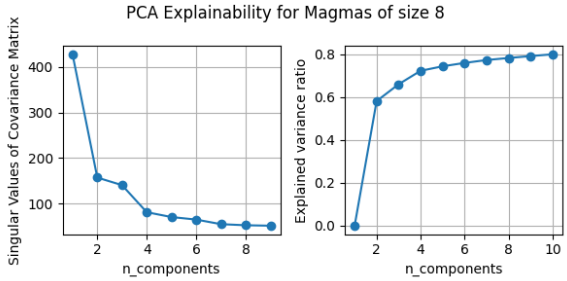

Figure 4: PCA Explainability statistics for a random sample of $n = 20\,000$ magmas of size 8. The values of the first ten singular values (left) and the explained variance ratio (right) for the first ten principal components.

choice of picking the three first dimensions of the PCA is justified by our visualization purposes as well as by the fact that a majority of variability in the data is captured by the first three principal components, as shown in Fig. 4.

6.2 The latent space is good at detecting signatures of equational theories

Exploring the latent space and looking at each position of the 4964 equational theories, we observe that the latent space is surprisingly good at identifying and grouping together the equational theories $\mathsf{Eqn}[k]$ with the same signature, see Fig. 5.

Here, by signature of an equational theory, we mean the ordered pair (a, b) describing the numbers a and b of diamonds on the left-hand side and right-hand side of the equation, respectively. Typically, the Tarski's law $\mathsf{Eqn}[542]$

$$x = (z \diamond (x \diamond (y \diamond z)))$$

has signature $(0, 3)$. We indicate how the 4694 equational theories are partitioned according to their signature, and how they are colored in the latent space depicted in Fig. 5.

2842	theories have signature	$(0,4)$	[red]
1015	theories have signature	$(1,3)$	[blue]
427	theories have signature	$(2,2)$	[green]
260	theories have signature	$(0,3)$	[purple]
104	theories have signature	$(1,2)$	[cyan]
30	theories have signature	$(0,2)$	[pink]
9	theories have signature	$(1,1)$	[yellow]
5	theories have signature	$(0,1)$	[pink]
2	theories have signature	$(0,0)$	[pink]

6.3 Expectation and variance

The *expectation* and *variance* of an equational theory $\mathsf{Eqn}[k]$ are defined as the expectation and variance of the associated sequence of Stone pairings:

$$E[\,\mathsf{Eqn}[k]\,] = \tfrac{1}{n}\textstyle\sum_{i=1}^{n} p_i$$
$$V[\,\mathsf{Eqn}[k]\,] = \tfrac{1}{n}\textstyle\sum_{i=1}^{n} \Big(p_i - E[\,\mathsf{Eqn}[k]\,]\Big)^2$$

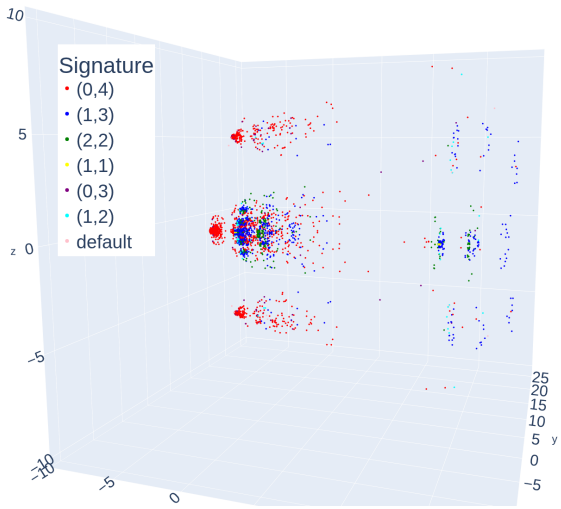

Figure 5: Latent space of equational theories where every theory is colored according to its signature (a, b), where a and b are the numbers of diamond operations on the lefthand and righthand side of the theory.

where $p_i = \langle\, \mathsf{Eqn}[k] \mid A_i \,\rangle$. Note that the expectation and variance do not depend on the sampling order of the finite magmas $A_1, \ldots, A_n$ and can be thus computed from the Stone spectrum of $\mathsf{Eqn}[k]$.

6.4 The X-axis is strongly correlated to expectation

The first principal component X of a theory $\mathsf{Eqn}[k]$ is strongly correlated to its expectation $E[\,\mathsf{Eqn}[k]\,]$, as shown by the linear regression graph, see Fig. 8 in Appendix. As a consequence, the theory with maximum X-value is $\mathsf{Eqn}[0]$

$$x = x$$

with expectation exactly 1, followed by $\mathsf{Eqn}[3277]$ and its conjugate $\mathsf{Eqn}[4067]$

$$x \diamond x = ((x \diamond x) \diamond y) \diamond y$$
$$x \diamond x = y \diamond (y \diamond (x \diamond x))$$

whose expectation in the sample of finite magmas of size 8 we took is around .46. On the other side of the latent space, one of the equational theories with smallest X-value is $\mathsf{Eqn}[1]$

$$x = y$$

with expectation exactly $\frac{1}{8} = .125$ in any finite magma of size $N = 8$ used in the sample.

6.5 The Y-axis is correlated to variance

The principal component Y of an equational theory $\mathsf{Eqn}[k]$ is strongly correlated to its variance $V[\,\mathsf{Eqn}[k]\,]$, as shown by the linear regression, see Fig. 9 in Appendix. The equational theory with minimum variance and Y-value is $\mathsf{Eqn}[4275]$ and conjugate $\mathsf{Eqn}[4590]$:

$$x \diamond (x \diamond x) \;=\; y \diamond (y \diamond y)$$
$$(x \diamond x) \diamond x \;=\; (y \diamond y) \diamond y$$

The equational theory with maximum variance and Y-value is $\mathsf{Eqn}[3658]$ which is self-conjugate:

$$x \diamond x \;=\; (x \diamond x) \diamond (x \diamond x)$$

6.6 The Z-axis detects conjugacy

One immediate and striking observation on the latent space is the Z-value produces a reflection symmetry between conjugate theories, around the plane $Z = 0$. We noticed this emerging reflection symmetry between conjugate theories in our first experiments, and improved this reflection symmetry by selecting a perfectly symmetric sample of finite magmas. For reference, the equational theory with maximum Z-value is $\mathsf{Eqn}[1022]$

$$x \;=\; x \diamond ((x \diamond (x \diamond y)) \diamond y)$$

closely followed by $\mathsf{Eqn}[428]$

$$x \;=\; x \diamond (y \diamond (x \diamond (y \diamond x)))$$

By reflection symmetry, the theory with minimum Z-value is the conjugate $\mathsf{Eqn}[2529]$:

$$x \;=\; (y \diamond ((y \diamond x) \diamond x) \diamond x$$

closely followed by the conjugate $\mathsf{Eqn}[3067]$:

$$x \;=\; (((x \diamond y) \diamond x) \diamond y) \diamond x$$

Accordingly, there are 84 self-conjugate equational theories and they all appear on the plane $Z = 0$. This ability of the Z-component to detect conjugacy is remarkable since two conjugate theories $\mathsf{Eqn}[j]$ and $\mathsf{Eqn}[k]$ have the same Stone spectrum, and thus the same expectation and variance, see the definition in §5.3 and discussion in §6.3.

7 Clustering of provably equivalent theories in the latent space

Now that we have constructed the latent space of equational theories, we can embed the implication graph $(G, \rightsquigarrow)$ to visualize what it looks like in three dimensions. For every pair of equational theories related by a proof $\mathsf{Eqn}[j] \Longrightarrow \mathsf{Eqn}[k]$ we thus draw an *edge* $\mathsf{Eqn}[j] \longrightarrow \mathsf{Eqn}[k]$ between the vertices associated to $\mathsf{Eqn}[j]$ and $\mathsf{Eqn}[k]$ in the latent space. We use the same terminology as for implications, and thus distinguish three classes of edges: the *reversible* edges $\sim$, the *atomic* edges $\rightsquigarrow$ which are strict by definition, and the *strict* edges $\Rightarrow$ which are general and not necessarily atomic. The length of an edge can be computed in several ways. In this paper, we choose the simplest distance which is the Euclidian distance in the latent space.

At this stage, it is important to notice that two provably equivalent theories $\mathsf{Eqn}[j]$ and $\mathsf{Eqn}[k]$ may induce different Stone pairings within a given finite magma $(A, \diamond)$. Indeed, $\mathsf{Eqn}[j] \sim \mathsf{Eqn}[k]$ simply ensures that validity is preserved

$$(A, \diamond) \vDash \mathsf{Eqn}[j] \iff (A, \diamond) \vDash \mathsf{Eqn}[k]$$

in other words, that the Stone pairing of $\mathsf{Eqn}[j]$ is equal to 1 if and only if the Stone pairing of $\mathsf{Eqn}[k]$ is equal to 1. On the other hand, we observe empirically that the reversible edges are on average seven times shorter than strict atomic edges, and ten times shorter than general stric edges, as indicated in the table below.

	Reversible	**Atomic**	**Strict**
count	2 470 916	1 052 209	5 707 363
mean	0.69	5.29	7.17
std	0.65	2.22	4.30
min	0.00	0.00	0.00
25%	0.00	4.14	4.17
50%	0.63	4.69	6.14
75%	0.96	6.33	9.22
max	5.36	37.08	41.20

This important difference suggests a degree of clustering of the provably equivalent theories in the latent space. The hypothesis can be validated by computing the center of mass of each clique of provably equivalent theories, and connecting each theory to the center of mass of its corresponding clique. One obtains in this way a three-dimensional picture, see Fig. 6, where the clusters of provably equivalent theories are clearly separated.

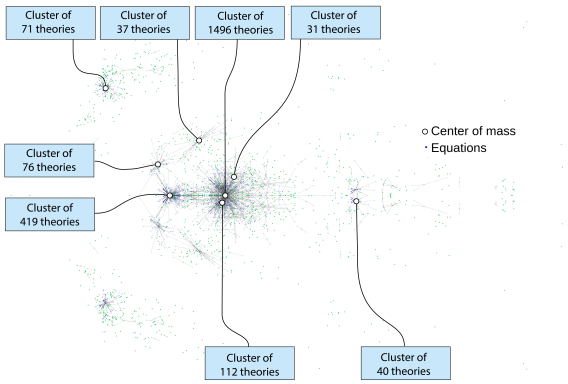

Figure 6: Visualizing the center of mass of each provably equivalent clique. Each equational theory is drawn connected to its corresponding center of mass.

8 Implication flows in the latent space

The clustering of provably equivalent theories in the latent space observed in §7 has an important